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Original Article

Construction of an InstantPC-derivatized glycan glucose unit database: A foundation work for high-throughput and high-sensitivity glycomic analysis

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The glycosylation profile of biotherapeutic glycoproteins is a critical quality attribute that is routinely monitored to ensure desired product quality, safety and efficacy. Additionally, as one of the most prominent and complex post-translational modifications, glycosylation plays a key role in disease manifestation. Changes in glycosylation may serve as a specific and sensitive biomarker for disease diagnostics and prognostics. However, the conventional 2-aminobenzamide-based N-glycosylation analysis procedure is time-consuming and insensitive with poor reproducibility. We have evaluated an innovative streamlined 96-well-plate-based platform utilizing InstantPC label for high-throughput, high-sensitivity glycan profiling, which is user-friendly, robust and ready for automation. However, the limited availability of InstantPC-labeled glycan standards has significantly hampered the applicability and transferability of this platform for expedited glycan structural profiling. To address this challenge, we have constructed a detailed InstantPC-labeled glycan glucose unit (GU) database through analysis of human serum and a variety of other glycoproteins from various sources. Following preliminary hydrophilic interaction liquid chromatography (HILIC) with fluorescence detection separation and analysis, glycoproteins with complex glycan profiles were subjected to further fractionation by weak anion exchange HILIC and exoglycosidase sequential digestion for cross-validation of the glycan assignment. Hydrophilic interaction ultra-performance liquid chromatography coupled with electrospray ionization mass spectrometry was subsequently utilized for glycan fragmentation and accurate glycan mass confirmation. The constructed InstantPC glycan GU database is accurate and robust. It is believed that this database will enhance the application of the developed platform for high-throughput, high-sensitivity glycan profiling and that it will eventually advance glycan-based biopharmaceutical production and disease biomarker discovery.

Key words: glycan GU database; glycomic analysis; HILIC-FLD; InstantPC; protein glycosylation.

Introduction

Protein glycosylation has been recognized as one of the most common, complex and ubiquitous post-translational modifications (PTM) (Spiro 2002; Moremen et al. 2012) and is involved in various physiological processes. In addition to being widely distributed in eukaryotic cells (Corfield 2017), glycoproteins and polysaccharides also have important biological functions in bacterial cells (Benz and Schmidt 2002; Abu-Qarn et al. 2008; Nothaft and Szymanski 2010) and most viruses (Sugrue 2007; Vigerust and Shepherd 2007; Watanabe et al. 2019). The biological functions of glycans can be broadly divided into four distinct categories that include structural and modulatory roles, extrinsic (interspecies) recognition, intrinsic (intraspecies) recognition and molecular mimicry of host glycans (Varki and Lowe 2009; Varki 2017). The cellular protein glycosylation process is highly sensitive to the biochemical environment and plays a key role in disease manifestation in which specific changes in glycan composition may be directly linked to disease. For this reason, monitoring changes in glycosylation may serve as a specific and sensitive approach to discover biomarkers for disease diagnostics and prognostics (Freeze 2013; Kirwan et al. 2015; Reily et al. 2019). Glycans themselves can be used as therapeutic molecules and this has given rise to an emerging field that has gradually gained extensive research interest to identify and exploit innovative glycan-based therapeutic strategies (Balzarini 2007; Hudak and Bertozzi 2014; Seeberger and Cummings 2015; Valverde et al. 2019; Costa et al. 2020). Additionally, in biopharmaceutical development, glycosylation of proteins such as monoclonal antibodies is well recognized as a critical quality attribute and is routinely monitored to within an appropriate limit, range or distribution to ensure the desired product quality, safety and efficacy (Reusch and Tejada 2015).

However, the diversity and complexity of glycan structures present huge analytical challenges for effective identification of glycan structures and integration to structure–function relationships. The continuous development of reliable, robust, reproducible and sensitive analytical methodologies for exploiting glycan structural diversity is critical to gain greater understanding of the physiological and pathological processes that are regulated by protein glycosylation. A variety of approaches are now available for determining the complete glycan structural heterogeneity, including monosaccharide composition, sequence, branching and glycosidic linkage. High-performance liquid chromatography (HPLC) and capillary electrophoresis are the key parts of many glycan analytical workflows, with hydrophilic interaction liquid chromatography coupled with fluorescence detection (HILIC-FLD) separation and analysis of 2-aminobenzamide (2-AB)-derivatized glycans being the standard method. Mass spectrometry (MS) has also been a predominant method for the characterization and elucidation of glycan structures

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following controlled proteolytic breakdown or enzymatic release of glycans from glycoproteins. Nuclear magnetic resonance (NMR) is an alternative and important method to determine the structure, dynamics, recognition and interaction of complex glycans. Microarray-based technology is emerging as a powerful new tool for the dynamic analysis of glycan expression (Li 2010; Everest-Dass et al. 2018). However, few methods have been developed for low sample consumption and for high-throughput glycomic analysis. The availability of high-throughput, high-sensitivity platforms for streamlined and standardized glycan profiling is essential for monitoring product characteristics in a bioprocess as well as for providing unique insights into the cellular expression of glycan expression under normal physiological conditions and in human disease.

In a previous publication, we presented a novel streamlined 96-well-plate-based glycan preparation platform for high-throughput, high-sensitivity glycan profiling with potential application to biopharmaceutical development and disease biomarker discovery (Xie et al. 2021). By taking advantage of rapid methods of glycoprotein denaturation, deglycosylation and glycan fluorescent derivatization, the whole process for glycan sample preparation from a variety of sources was shortened significantly to only 10 min. Additionally, with the use of the rationally designed innovative fluorescent dye, InstantPC, the workflow was highly sensitive for both FLD and MS detection. The sensitivity of the method allows the detection of low abundant glycan species which were previously undetectable by alternative methods. However, the commercial availability of InstantPC-labeled glycan standards is limited, which makes it difficult to accomplish accurate glycan characterization and quantification without the help of sophisticated instrumentations such as MS or NMR. This is a major drawback to the developed platform as a rapid analytical method for monitoring glycoproteins.

To mitigate this difficulty, we have constructed a detailed glucose unit (GU) database for InstantPC-labeled glycans monitored by HILIC-FLD method. A variety of glycoproteins from various sources have been investigated: human serum, lactoferrin from bovine milk, fetuin from fetal bovine serum, human transferrin, ovalbumin from chicken egg white, ribonuclease A type III-A from bovine pancreas and a monoclonal antibody (EG2-hFc) produced from Chinese hamster ovary (CHO) cells. The selection of these glycoproteins was made with the intention of including three different classes of glycans: high mannose, complex and hybrid structures, which are commonly attached to biotherapeutics or being investigated for glycan-based biomarker discovery from serum and plasma. The glycan samples were first prepared by following the protocol of the high-throughput, high sensitivity workflow as described previously (Xie et al. 2021), followed by HILIC-FLD separation. For the more complex glycan profiles, an extra step of weak anion exchange (WAX) hydrophilic interaction liquid chromatography (WAX-HILIC) fractionation was employed to separate the charged fractions prior to HILIC analysis, exoglycosidase sequential digestion and glycan assignment. Hydrophilic interaction ultra-performance liquid chromatography coupled with electrospray ionization mass spectrometry (HILIC-UPLC-ESI-MS) was further utilized for glycan fragmentation and accurate glycan mass verification.

The constructed InstantPC-labeled glycan GU database contains the values for both high- and ultra-performance HILIC-FLD (H(U)PLC-HILIC-FLD). They were extensively screened using different liquid chromatographic systems, different HILIC columns, different column temperatures and even different liquid phase elution gradients. Finally, human serum polyclonal immunoglobulin G (IgG) was selected to cross-validate the constructed InstantPC glycan GU database, and the reported GU database is accurate and robust with good reproducibility and applicability. It is believed that this detailed InstantPC glycan GU database will significantly improve the applicability of the developed streamlined glycan profiling platform with minimal effort and lay a solid foundation for advancement in glycan-related biopharmaceutical development and biomedical research. The InstantPC glycan GU database will be made accessible in the GlycoStore website (https://glycostore.org/) (Zhao et al. 2018).

Results

Strategies for construction of InstantPC-labeled glycan GU database

HILIC-FLD followed by GU value normalization against related fluorophore-labeled dextran ladder has been used routinely as a robust and reliable way for glycan analysis (Guile et al. 1996; Melmer et al. 2010; Hilliard et al. 2017; Zhao et al. 2018). Compared to other chromatographic techniques employing an internal standard for retention time normalization (Pabst et al. 2007), the potential analytical variables, such as different experimental settings, HILIC columns, column temperatures and elution gradient, are reduced or minimized significantly. Figure 1 shows the workflow that was used in constructing the InstantPC-labeled glycan GU database. The upper panel shows the various methods that were used in generating the experimental data used to support the database. This included WAX-HILIC glycan fractionation, exoglycosidase sequential digestion, HILIC-FLD glycan separation and HILIC-UPLC-ESI-MS glycan mass determination. Selected glycoproteins were used to generate experimental data using combinations of these methods.

The database was developed in multiple stages. The first stage involved glycan profiling of each glycoprotein by HILIC-FLD using high-throughput glycan sample preparation following the previously reported 96-well-plate platform (Xie et al. 2021). Structural assignments of the glycan profiles were validated further by reference to published profiles using other labels such as 2-AB. For the second stage, WAX-HILIC with FLD (WAX-HILIC-FLD) was employed to fractionate the InstantPC-labeled glycans for the complex profiles that were considered to include multiple charge species. In the third stage, the labeled glycans were subjected to extensive exoglycosidase sequential digestion to enable detailed glycan structural assignment. This was applied to samples following HILIC-FLD separation or collected fractions following WAX-HILIC-FLD. The reaction array of exoglycosidase sequential digestion is a method that has been used extensively to deduce glycan structures from the known specificity of breakdown of the glycosidases used (Guile et al. 1996). In the fourth stage, HILIC-UPLC-ESI-MS was used to determine the exact molecular mass of each glycan to validate the structural assignments deduced by chromatography.

The resulting InstantPC glycan GU database is shown in supporting information Supplementary Table SI. This has 264

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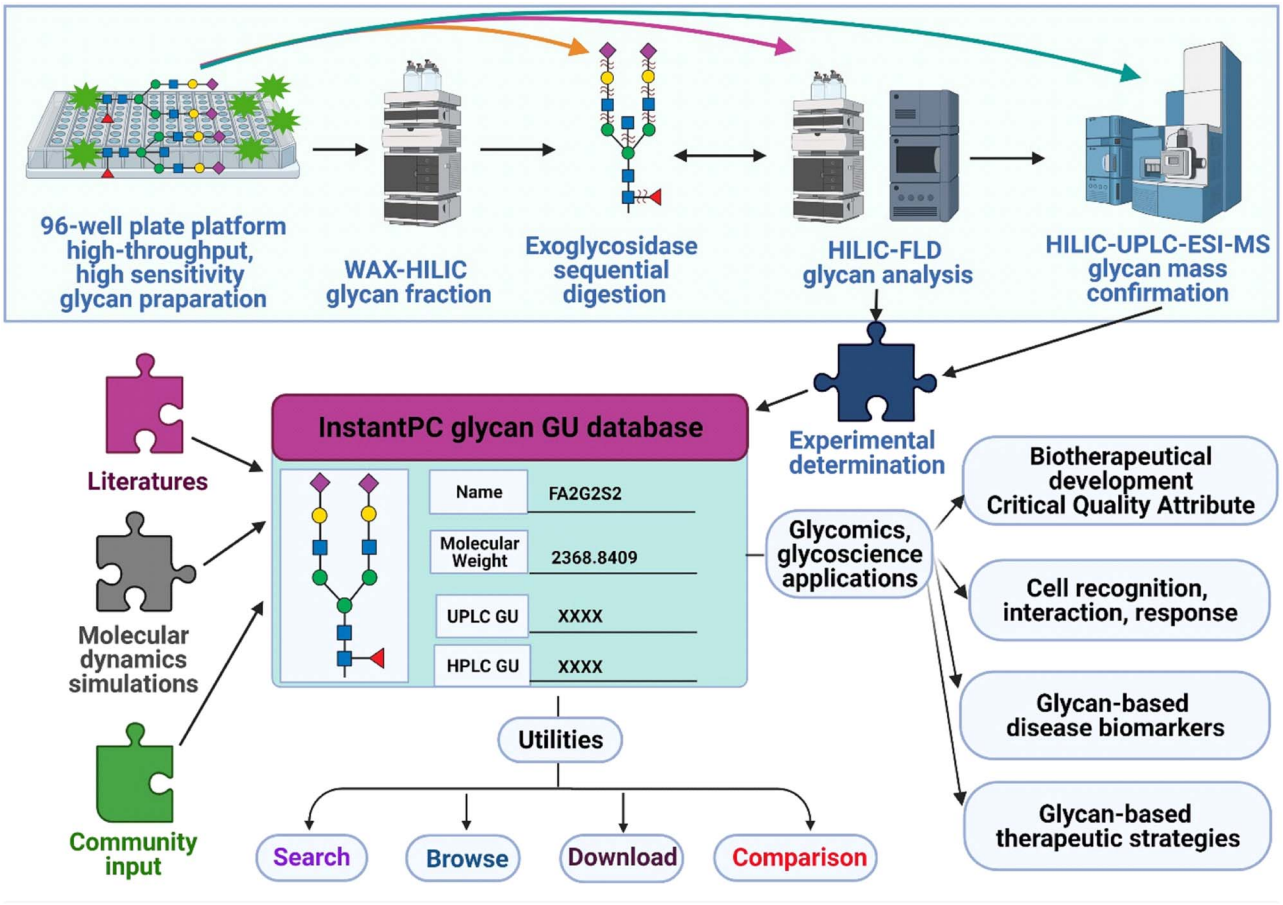


Fig. 1. Strategies for construction of an InstantPC glycan GU database, mainly involving experimental determination, literatures review, MD simulations and future community input. The experimental determination includes high-throughput glycan sample preparation, WAX-HILIC glycan fractionation, exoglycosidase sequential digestion, HILIC-FLD glycan separation and HILIC-UPLC-ESI-MS glycan mass determination. The constructed InstantPC glycan GU database can be used for search, browse, download and comparison, with potential application in glycomics or glycoscience. Created with BioRender.

unique entries of glycans analyzed from our selected recombinant and serum glycoproteins. Each glycan entry is assigned an accession ID (GlyTouCan) (Aoki-Kinoshita et al. 2016; Tiemeyer et al. 2017; Fujita et al. 2021) and is represented by its symbol nomenclature for graphical representation (SNFG) (Varki et al. 2015; Neelamegham et al. 2019). This is followed by the molecular weight as determined by the MS and GU values based upon HPLC and UPLC. The molecular weights of the entries vary within the range of 586–3973. The GU values from HPLC analysis range between 1.8 and 17.3, whereas the UPLC values are slightly different with a range from 1.7 to 17.4. The database covers the three major types of glycan structures: complex (211), hybrid (32), high mannose (17) as well as paucimannose structures (4 as $Man_{<5}GlcNAc_2$). Other notable structural features in the database include 189 sialylated, 75 neutral, 90 core fucosylated and 23 outer-arm fucosylated glycans. Of the complex glycans, 124 are biantennary, 99 are triantennary and 34 are tetraantennary, with 53 classified as bisecting glycans.

2D-LC with exoglycosidase sequential digestion for glycan structural determination

To generate the InstantPC-labeled glycan GU database, accurate and unbiased structural assignments of chromatographically separated peaks were required. However, due to the intrinsic diversity and complexity of glycan structures, overlapping and superimposed chromatographic peaks are routinely encountered for glycoproteins with complex glycan profiles. This complicates glycan structural assignments even with the help of exhaustive exoglycosidase sequential digestion. To overcome this problem, we employed 2D liquid chromatography (2D-LC) for glycan fractionation, separation, detection and characterization. This offered significant improvement in resolving power over conventional 1D liquid chromatography (1D-LC). The InstantPC-labeled glycans were fractionated by WAX-HILIC-FLD in the first-dimension liquid chromatography based on the glycan nature of charge, size and polarity, and subsequently, the collected glycan fractions were separated by HILIC-FLD in the second-dimension liquid chromatography based on the glycan nature of hydrophilicity (or polarity) only.

An example of the use of the 2D-LC technique is shown in Figure 2 for the analysis of human serum. Figure 2A and B show the independent WAX-HILIC-FLD and HILIC-FLD glycan profiles, respectively, obtained from human serum. The glycan profiles are complex, spanning a wide range of chromatographic retention times, making it impossible for accurate glycan assignment due to multiple overlapping peaks. Exoglycosidase sequential digestion of the HILIC separation was attempted but was difficult to interpret because of further

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overlapping peaks (data not shown). Therefore, an extra dimension of LC fractionation to simplify the complex glycan profiles turned out to be necessary. As shown in Figure 2A, a total of five distinct glycan fractions were obtained by WAX-HILIC-FLD fractionation, S0, S1, S2, S3 and S4, representing the fraction of neutral glycans and glycans containing 1, 2, 3 and 4 sialic acid residues, respectively. Subsequently, each glycan fraction was subjected to further HILIC-FLD separation. This simplified the glycan profiles substantially.

Figure 2C shows the stacked HILIC-FLD chromatographs of fractions S0–S4 from the initial WAX-HILIC-FLD separation, with the chromatographic superimposition of all the glycan fractions representing the whole and unfractionated glycan profiles for human serum (Figure 2B). Exhaustive exoglycosidase sequential digestion for each glycan fractions was then carried out for glycan structural assignment. As an example, Figure 2D1–D7 shows a representative set of results of the WAX-HILIC-FLD S2 fraction subjected to an array of exoglycosidases to cleave α2-3 sialic acid, α2-3,6 sialic acid, galactose, GlcNAc and α1-6 core fucose sequentially. With significantly simplified glycan profiles for each fraction, unbiased glycan assignment became possible.

Compared to size and polarity, glycan charge plays a dominant role in WAX-HILIC-FLD fractionation and therefore is suitable for samples which are heavily sialylated and show glycan species with different charges. Other glycan profiles are more limited in their content of charge variants and have predominately neutral glycans (no charges). This is the case for some of the glycoprotein analysis shown in the supporting information (Supplementary Figures S1–S7) in which profiles were obtained using HPLC and UPLC chromatography. For example, analysis of ovalbumin and lactoferrin showed that the majority of the glycans were neutral and so the benefit of WAX-HILIC-FLD fractionation is more limited (supporting information Supplementary Figures S3 and S4).

Glycan structural confirmation by HILIC-UPLC-ESI-MS

Structural assignments by chromatography were validated by MS. In our system, HILIC-UPLC was coupled to ESI-MS (HILIC-UPLC-ESI-MS) for glycan fragmentation and accurate glycan molecular mass determination. An example of this analysis is shown in Figures 2 and 3. The glycan peak labeled with a red star in Figure 2B or D-1 is shown as an example of the type of MS fragmentation obtained. Figure 3 shows the ion m/z of 1242.97 as the base peak, and this indicated a possible glycan structural assignment as A2G2S2 based on a search of the GlycoWorkbench database. The molecular mass for A2G2S2 is 2222.783, and the molecular mass addition of InstantPC dye is 261.148. ESI-MS detection was in the positive mode with molecular mass addition of one proton as 1.008. Therefore, the theoretical ion m/z for A2G2S2-InstantPC was calculated to be (2222.783 + 261.148 + 1.008)/1 =2484.939 and (2222.783 + 261.148 + 1.008 + 1.008)/2 = 1242.974 at charge of 1H and 2H, respectively. Thus, the calculated theoretical ion m/z values were in good agreement with the measured values by ESI-MS. Additionally, the glycan fragmentation patterns further confirmed that the investigated glycan structure was A2G2S2 (Figure 3).

For glycoproteins with simpler glycan profiles, 2D-LC fractionation was not necessary, and sufficient data could be provided by HILIC-FLD separation alone followed by exoglycosidase sequential digestion. An example is shown, in Figure 4, of the profiles obtained from ribonuclease A. As shown in Figure 4A and B, ribonuclease A consists of a variety of mannose containing glycans, and accurate glycan assignment was achieved by using only HILIC-FLD separation and mannosidase digestion. Further analysis by HILIC-UPLC-ESI-MS method enabled the confirmation of the assigned glycan structures. Figure 4C shows as an example how the glycan peak labeled with a red star in Figure 4A was analyzed and assigned as Man9. As shown in Figure 4C, the glycan displayed a more complex MS spectrum than the above example of A2G2S2. The peak with an ion m/z of 1072.90 was the base peak, and the possible glycan structure was Man9 according to database searching. The molecular mass for Man9 is 1882.645. Therefore, the theoretical ion m/z for Man9-InstantPC was calculated to be (1882.645 + 261.148 + 1.008)/1 = 2144.801 and (1882.645 + 261.148 + 1.008 + 1.008)/2 = 1072.905 at charge of 1H and 2H, respectively. The calculated theoretical ion m/z values were in good agreement with the measured values by ESI-MS. Additionally, the glycan fragmentation patterns further confirmed that the investigated glycan structure was Man9 (Figure 4C).

Verification of the InstantPC glycan GU database

As shown in supporting information, Supplementary Table SI, the constructed InstantPC glycan GU database consisted of a variety of glycans spanning a wide range of GU values from common glycoproteins found in biopharmaceutical products and biological fluids. The database was generated for both UPLC and HPLC systems, which was deemed necessary for two major reasons. One reason was the experimental deviations of the determined GU values for identical glycan structure varied significantly between the UPLC and HPLC systems compared to other experimental factors, such as HILIC column, column temperature or elution gradient. The other reason was to extend the database to any glycoscience-focused analytical laboratory equipped with basic HPLC system where routine accessibility to more expensive UPLC system is limited.

To further test the applicability and robustness of the constructed database, we applied the whole workflow on a well-studied polyclonal antibody, human IgG. The glycans were prepared from IgG by following the developed high-throughput workflow and were subsequently analyzed by the HILIC-FLD method using both UPLC and HPLC systems. After retention time normalization against the InstantPC-derivatized dextran ladder, the GU values for each glycan were obtained. Our constructed InstantPC glycan GU database was then searched for glycan structure assignments. The structural assignments for the InstantPC-labeled glycans from IgG were then further verified by HILIC-UPLC-ESI-MS.

Figure 5 shows the HILIC-FLD separations of the IgG glycans alongside dextran ladders by HPLC and UPLC. It was not surprising that UPLC demonstrated superior separation and sharper glycan peaks than HPLC. However, the glycan profiles were similar despite the use of different LC systems and HILIC columns for the analysis. However, the retention time for each glycan was significantly different between LC systems. This showed the necessity for glycan GU normalization against an InstantPC dextran ladder. As shown in Table I, the GU value for each InstantPC-labeled

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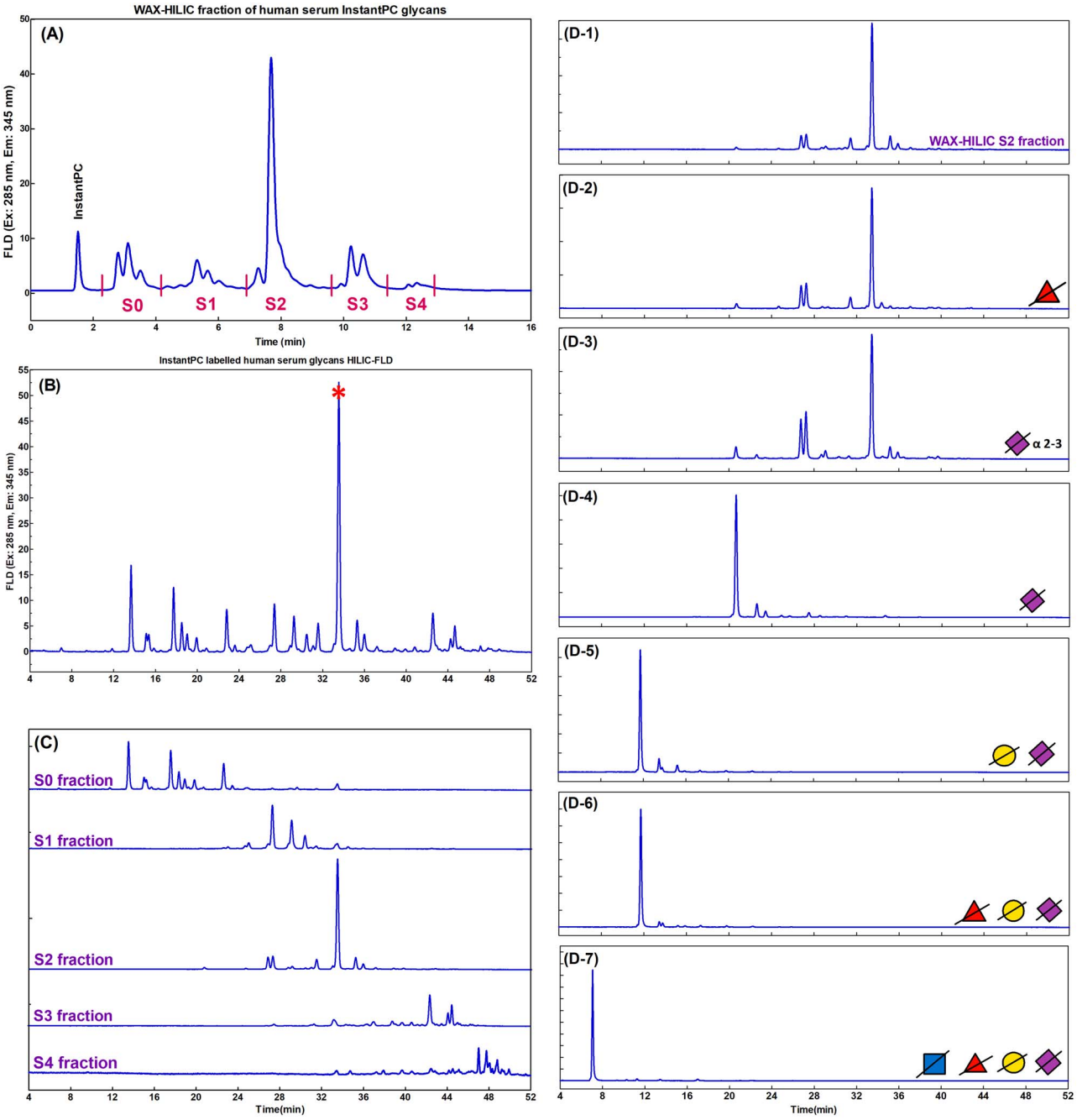


Fig. 2. Representative workflow for structural assignment of InstantPC-labeled *N*-glycans from human serum. (**A**) Chromatogram for WAX-HILIC-FLD fractionation of InstantPC-labeled *N*-glycans from human serum according to glycans charges (number of sialic acids), size and polarity with fractions S0, S1, S2, S3 and S4 representing glycans possessing 0, 1, 2, 3 and 4 sialic acids, respectively. (**B**) HILIC-FLD chromatogram for the total unfractionated profile of InstantPC-labeled glycans from human serum. (**C**) HILIC-FLD chromatograms for each WAX-HILIC-FLD fraction of S0, S1, S2, S3 and S4, respectively. (**D**) Representative exoglycosidase sequential digestion array for WAX-HILIC-FLD S2 fraction of InstantPC-labeled glycans from human serum analyzed by HILIC-FLD. α (1-2,4,6)-fucosidase O (Omnitrophica bacterium) to cleave α1-6 core fucose; sialidase S (*S. pneumoniae*) to cleave α2–3 sialic acid; sialidase a (*A. ureafaciens*) to cleave α2-3,6,8,9 sialic acid; β (1-3,4)-galactosidase (bovine testis) to cleave galactose and β-*N*-acetylhexosaminidase (*S. pneumoniae*) to cleave GlcNAc, respectively. Symbols: blue square: *N*-acetylglucosamine; yellow circle: galactose; red triangle, fucose; purple diamond: *N*-glycolylneuraminic acid. The peak labeled in red in fig. B or D-1 was selected to illustrate the glycan mass confirmation by HILIC-UPLC-ESI-MS in Figure 3.

glycan was calculated against the accompanied standard curve generated from the dextran ladder. The corresponding glycan structure was then obtained from a search of the InstantPC glycan GU database. The structural assignments were in good agreement with those for human IgG reported in the literature (Lauc et al. 2013; Trbojevic Akmacic et al. 2015; Theodoratou et al. 2016). Additionally, the exact molecular mass for each glycan was further verified by HILIC-UPLC-ESI-MS and demonstrated significant accuracy (data not shown). This showed that the constructed InstantPC glycan GU database is accurate and robust and can be applied to the analysis of other hitherto uncharacterized glycan samples in other analytical laboratories without major modifications of the experimental conditions.

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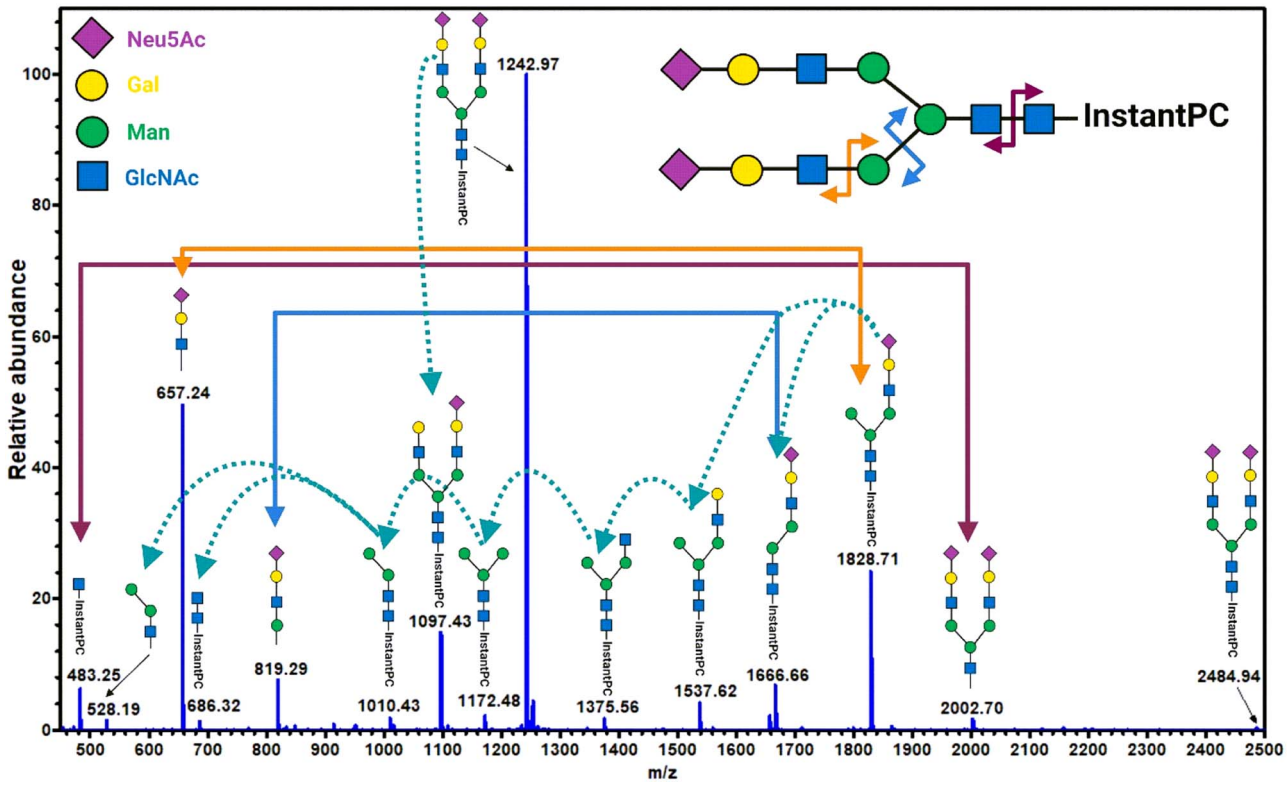


Fig. 3. Representative MS spectrum of InstantPC-labeled A2G2S2 to demonstrate the accurate glycan mass verification by HILIC-UPLC-ESI-MS. The glycan fragments and accompanied ion m/z values were labeled.

Table I. The retention time, GU values and glycan structures for InstantPC-labeled IgG glycans

Peak NO	UPLC-FLD		HPLC-FLD		Glycan structures
	RT (min)	GU value	RT (min)	GU value	
1	11.723	5.375	8.421	5.465	A2
2	13.504	5.857	9.848	5.995	FA2
3	15.216	6.296	11.329	6.489	FA2B, A2[6]G1, M5
4	15.675	6.412	11.655	6.592	A2[3]G1
5	16.344	6.578	12.284	6.785	A2BG1
6	17.546	6.876	13.210	7.060	FA2[6]G1
7	18.327	7.069	13.946	7.271	FA2[3]G1
8	18.89	7.207	14.427	7.406	FA2[6]BG1
9	19.653	7.396	15.178	7.614	FA2[3]BG1
10	20.682	7.650	15.990	7.834	A2G2
11	21.796	7.926	17.032	8.114	A2BG2
12	22.627	8.133	17.721	8.297	FA2G2
13	23.467	8.343	18.551	8.517	FA2BG2
14	25.006	8.728	19.746	8.834	FA2[3]G1S1, FA2[6]BG1S1
15	26.73	9.161	21.473	9.296	FA2[3]BG1S1
16	27.284	9.300	21.846	9.396	A2G2S1
17	29.114	9.758	23.507	9.848	FA2G2S1
18	30.385	10.074	24.803	10.205	FA2BG2S1
19	33.487	10.842	27.527	10.968	A2G2S2
20	34.362	11.059	28.436	11.226	A2BG2S2
21	35.199	11.267	29.227	11.451	FA2G2S2
22	35.917	11.447	29.917	11.648	FA2BG2S2

$GU\ (UPLC) = -0.2636 + 0.8527T - 0.05102T^2 + 0.002091T^3 - 0.00004165T^4 + 0.0000003234T^5.$

$GU\ (HPLC) = 0.2098 + 0.9638T - 0.05571T^2 + 0.002136T^3 - 0.00003933T^4 + 0.0000002828T^5.$

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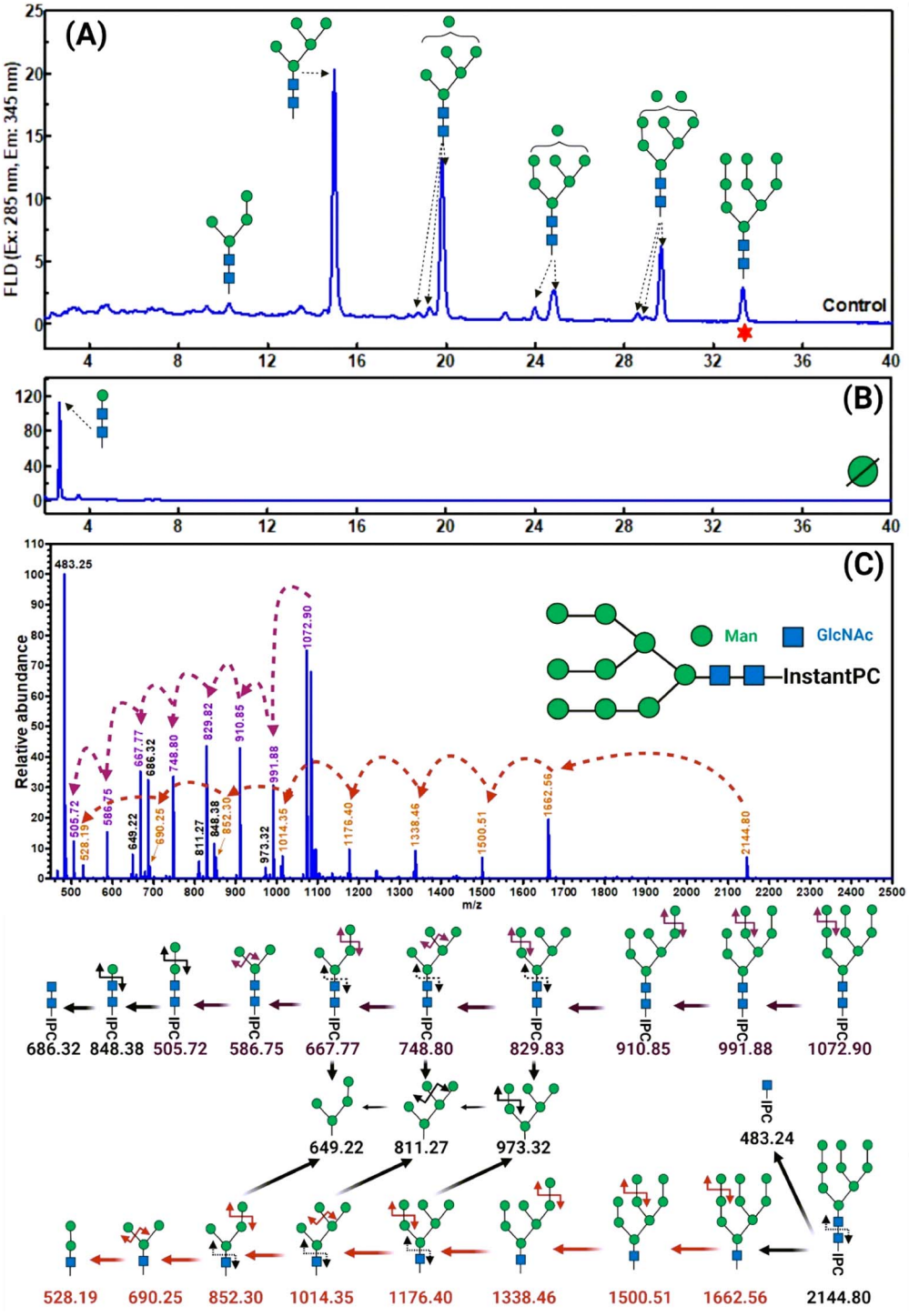


Fig. 4. Representative workflow for structural assignment and cross-validation of InstantPC-labeled glycans from bovine pancreas ribonuclease A. (**A**) HILIC-FLD chromatogram of InstantPC-labeled glycans from ribonuclease A. (**B**) HILIC-FLD chromatogram of InstantPC-labeled glycans from ribonuclease A after incubation with α (1-2,3,6)-mannosidase from Jack beans to cleave all α (1-2,3,6)-linked mannose residues to give rise to core trisaccharide structure of ManGlcNAc2 for cross-validation of glycan structure assignment. (**C**) Representative MS spectrum of InstantPC-labeled Man9 (labeled in red in **A**) to demonstrate the accurate glycan mass verification by HILIC-UPLC-ESI-MS. The glycan fragments and accompanied ion m/z values were labeled.

Core glycan and GU value increments for prediction of InstantPC glycan structures

Since glycans are highly hydrophilic, HILIC-FLD has currently been the method of choice for detailed protein glycan profiling. The HILIC separation and elution order of glycans are based on polarity (hydrophilicity) and usually the molecular size of the oligosaccharides, with additional selectivity for certain different monosaccharide and glycosidic linkages (Srebalus Barnes and Lim 2007). Theoretically, the structures of unknown glycans may be estimated or predicted by their HILIC-based GU values that are based on the retention times normalized against the corresponding

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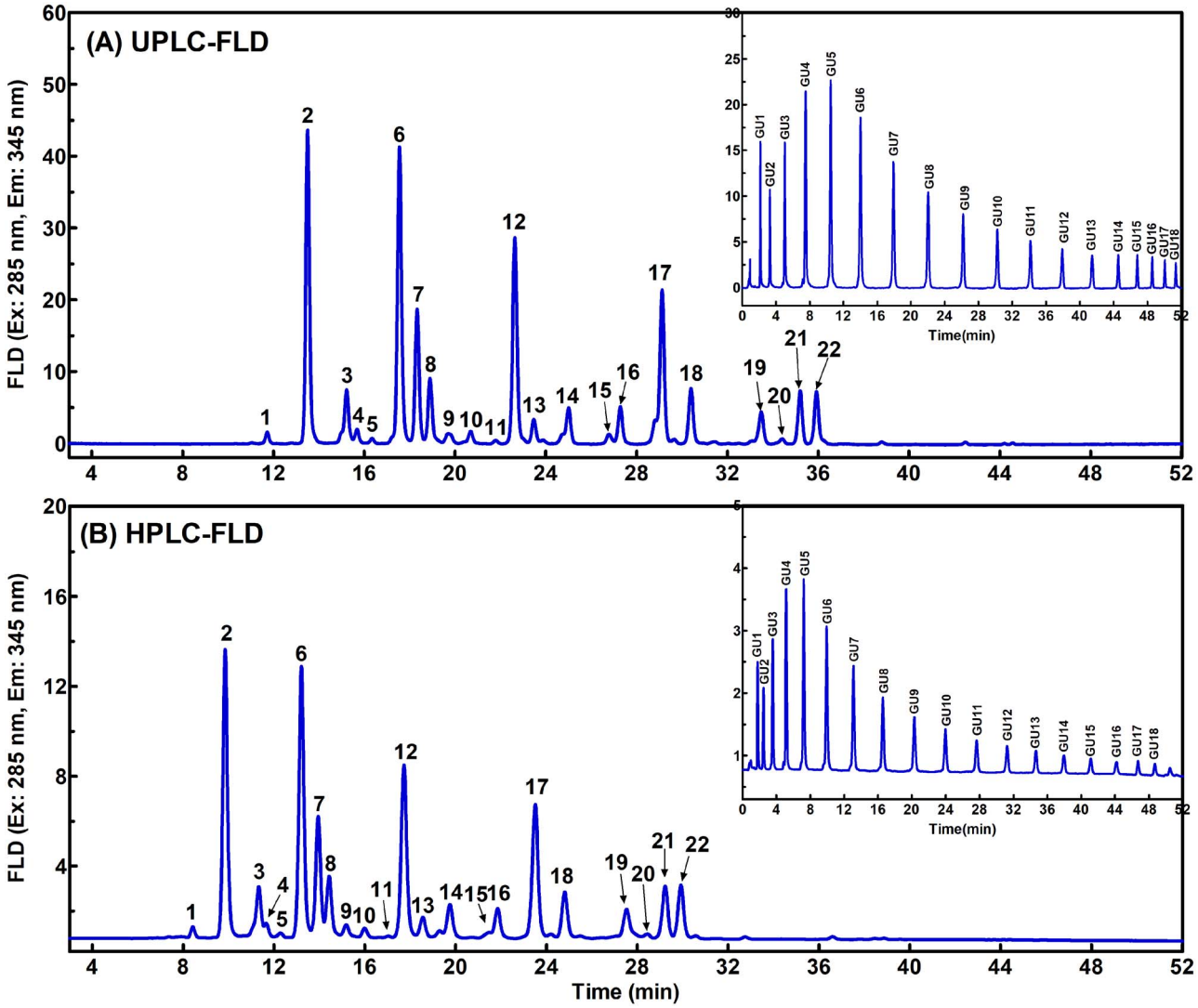


Fig. 5. HILIC-FLD chromatograms for cross-validation of the constructed InstantPC-labeled glycan GU database with known glycoprotein of human IgG separated by (**A**) UPLC and (**B**) HPLC under identical chromatographic conditions as described in the experimental section, with the inserts representing the InstantPC-labeled Maltodextrin calibration ladder, respectively.

fluorophore-labeled dextran ladder. This determination can be made from the core glycan structure Man3 GU value and the incremental GU values for adding specific monosaccharides to the related glycans.

Figure 6 shows a selective GU value panel for some representative glycan structures derived from the separation and analysis of InstantPC-labeled human serum. From left to right, starting from glycan core structure of Man3, with the addition of one monosaccharide sequentially, glycans of A1, A2, A2G1, A2G2, A2G2S1, A2G2S2 and their sibling glycans containing bisecting GlcNAc and α 1-6 core fucose can be obtained in this array. Accordingly, the GU values for the related glycans increase incrementally. It was clear that the additions of GU values to the glycan core structure and the subsequent glycan structures represented the correlation of specific monosaccharide extension to the core structure or the outer arms. For example, as shown in the top panel of Figure 6, the GU value for Man3 was determined to be 3.806 by sequentially adding GlcNAc(s), galactose(s) and sialic acid(s); the GU values for glycans of A1, A2, A2[3]G1, A2G2, A2G2S[3]1 and A2G2S2 were determined to be 4.586, 5.334, 6.387, 7.635, 9.294 and 10.856, respectively. These represented the GU value increments for GlcNAc at 0.780 and 0.748, galactose at 1.053 and 1.248, sialic acid at 1.659 and 1.562, respectively. Additionally, the GU value increment for core fucose was calculated to be 0.490 when adding one fucose to A2 to achieve FA2. However, the GU value increment for bisecting GlcNAc was the least significant compared to its counterparts, which was calculated to be only 0.306 when adding one bisecting GlcNAc to A2 to achieve A2B.

The chart in Figure 6 allows glycan structures to be predicted from the glycan core structure GU, the InstantPC monosaccharide GU incremental values, glycosidic linkages and arm specificity. This method of structural prediction was reported previously for 2-AB-labeled glycans (Guile et al. 1996), and our findings demonstrate that this can also be applied to InstantPC-labeled glycans. The different GU value increments for each monosaccharide addition are due to the significant structural difference between 2-AB and InstantPC. The nonreactive alkyl chain and the tertiary amine group possessed by InstantPC is absent from 2-AB, which makes the former more hydrophilic. This affects the HILIC elution

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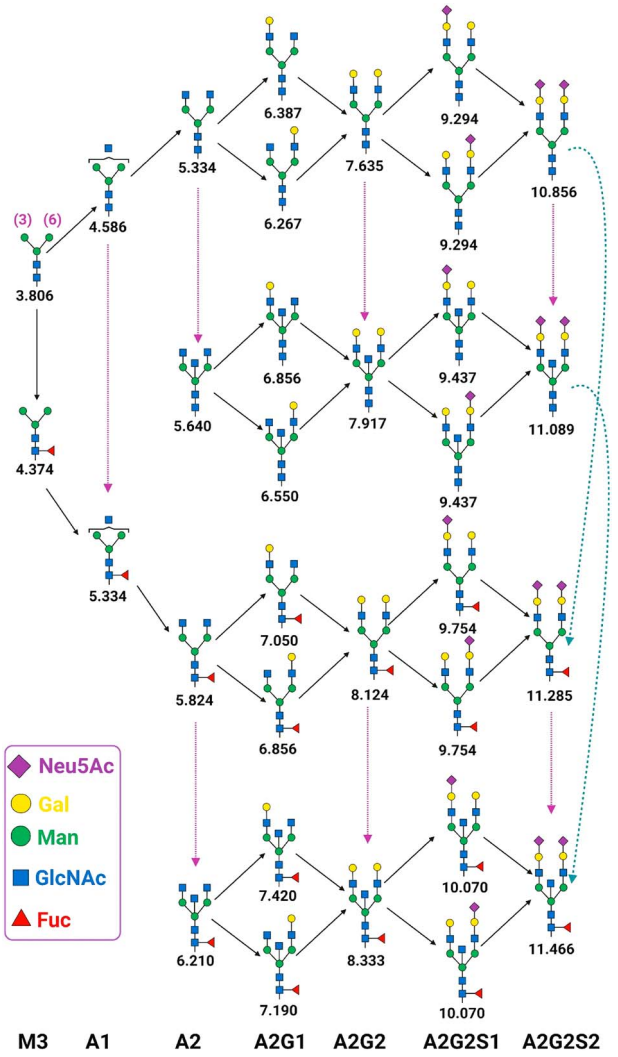


Fig. 6. Representative chart demonstrating the InstantPC-labeled glycan GU incremental values for the sequential addition of monosaccharide residues to M3. The values were determined from the GU values of the InstantPC-labeled glycans generated during the exoglycosidase sequential digestion of human serum.

time and consequently GU values significantly. Continuous investigation and comparison of GU value increments for different monosaccharides for a wider range of glycan structures under different HILIC-FLD conditions are deemed necessary to achieve more accurate prediction of GU values and unknown glycan structures. This is beyond the scope of the current research.

Accessibility and future contribution of the InstantPC glycan GU database

The generated InstantPC glycan GU database (Supplementary Table SI) covers a wide range of glycan structures, including high mannose, hybrid and complex glycoforms. It will be made available and generally accessible by uploading to the online glycan GU database GlycoStore (https://glycostore.org/) (Zhao et al. 2018). The GlycoStore online resource is searchable and downloadable, with potential matching glycan structures displayed following user input of GU values for a specific type of analysis. We anticipate that new glycan species will be discovered and identified in the future with the use of the InstantPC glycan analysis. Incorporating such data into an expanding InstantPC database input can be made possible on the website (https://glycostore.org/contribute) by uploading the experiment data in compliance with the minimum information required for a glycomics experiment (MIRAGE) guideline (York et al. 2014; Campbell et al. 2019). The MIRAGE is an initiative created by experts in the fields of glycobiology, glycoanalytics and glycoinformatics to design guidelines that improve the reporting and reproducibility of glycoanalytical methods.

GlySpace Alliance (http://www.glyspace.org) (Aoki-Kinoshita et al. 2020) is an ongoing international open access glycoinformatic initiative integrated GlyGen (York et al. 2020), Glycomics@ExPASy (Mariethoz et al. 2018) and GlyCosmos (Yamada et al. 2020) with the aim of promoting a standardized glycoinformatics infrastructure for effective and efficient communication within the glycobiology community. The current InstantPC glycan GU database implements the latest GlyTouCan registry Accession ID (Tiemeyer et al. 2017; Fujita et al. 2021) and symbol nomenclature for glycans (SNFG) notation (Neelamegham et al. 2019) for each identified glycan structure (as shown in Supplementary Table SI). Therefore, it can be easily integrated into the current ongoing international open access glycoinformatics initiative to support the development and extension of the glycan structure knowledgebase.

Discussion

Rapid sample preparation and increased sensitivity of analytical methods play a vital role in the advance of glycoscience or glycomic analysis either in the field of biopharmaceutical development or biomedical research. However, the conventional workflow for glycan sample preparation and analysis in many laboratories is time-consuming, labor-intensive, with low sensitivity and poor reproducibility. In our previous research, we have presented an innovative 96-well-plate-based platform for glycan sample preparation and analysis. The developed platform is high-throughput, with the whole glycan sample preparation process shortened significantly to only 10 min. The analytical method is highly sensitive and allows increased visibility of low abundant glycan species. Additionally, the developed platform is straightforward, user-friendly and can be adapted to any laboratory with minimal capital investment (Xie et al. 2021). However, due to the limited availability of InstantPC-labeled glycan standards and the lack of a reference GU database, the determination and structural assignment of InstantPC-labeled glycans can be a bottleneck. Therefore, the construction of a detailed InstantPC glycan GU database serves an important need to promote this analytical method.

Despite displaying different HILIC-FLD retention times, InstantPC-labeled glycans share significantly similar profiles to those labeled with the widely used 2-AB dye (Xie et al. 2021). The existence of an extensive 2-AB-labeled glycan GU library and database has been a valuable resource for identification of novel glycan profiles (Zhao et al. 2018). Our objective in this paper is to establish a similar database for InstantPC-labeled glycans. Our initial approach was to take advantage of the extensive literature of glycan profiles to select well-characterized glycoproteins that could be analyzed to form the basis of an InstantPC glycan GU database. A

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variety of glycoproteins from different sources was chosen for investigation, which included human serum (Saldova et al. 2014; Haakensen et al. 2016), ovalbumin from chicken egg white (Harvey et al. 2000; Yuan et al. 2014; Wang et al. 2018), lactoferrin from bovine milk (Nwosu et al. 2012; Figueroa-Lozano et al. 2020; Valk-Weeber et al. 2020; Zlatina and Galuska 2021), fetuin from fetal bovine serum (Sun et al. 2017), transferrin human partially iron saturated (Butler et al. 2003; Quaranta et al. 2016), ribonuclease A type III-A from bovine pancreas (Peracaula et al. 2003; Prien et al. 2009) as well as the monoclonal antibody of EG2-hFc produced from CHO cells studied in our previous research (Xie et al. 2021). The selection of glycoproteins was intended to cover the three classes of *N*-glycans: high-mannose, complex and hybrid structures, with a broad range of GU values. The selected glycoproteins provided an extensive coverage of the glycans derived from different species, which are commonly observed in biopharmaceutical products and as disease biomarkers.

The use of a 2D-LC method for the construction of the database was important. This offers great improvement in glycan fractionation, separation, detection and resolution over the conventional single dimension liquid chromatography (1D-LC). Fractionation by the first dimension was with an innovative mixed-mode surface chemistry column that combined both WAX and hydrophilic interaction (HILIC) retention mechanisms for optimal selectivity and resolution. The WAX functionality provides retention and selectivity for negatively charged glycans (mainly sialylated structures), while HILIC mode facilitates the separation of glycans of the same charge according to size and polarity. WAX-HILIC facilitates superior glycan fractionation compared to conventional WAX method where each fraction is difficult to distinguished clearly (Saldova et al. 2014). However, the high molecular size of the InstantPC-labeled glycans as well as the presence of hydrophilic nonreactive alkyl and tertiary amine groups makes the separation of peaks more complex compared to conventional 2-AB-labeled glycans (Thermo Scientific GlycanPac AXH-1 columns product manual). The second dimension used was HILIC-FLD where separation is based on the hydrophilicity (or polarity) of the glycan.

To the best of our knowledge, this is the first time that a 2D-LC method has been employed for the analysis of 6-aminoquinolyl-*N*-hydroxysuccinimidylcarbamate type fluorophores, including InstantPC, derivatized glycan separation and structural determination. The 2D-LC offers several benefits. (1) Significantly simplified profiles can be obtained for each glycan fraction and severe peak overlapping is minimized. (2) It is very flexible. After the first-dimension fractionation, the collected glycan fractions can be either subjected to the second-dimension separation or to exoglycosidase sequential digestion if necessary. (3) Extensive screening by exoglycosidase sequential digestion panel is not necessary since the glycans are grouped based on charge and the profiles for each group are simpler than for the whole glycan pool. (4) Potential time spent on the subsequent glycan structural determination is shortened since the overlapping of peaks is minimized. (5) The accuracy in glycan assignment is improved since sialylated glycan isomers can be distinguished easily.

Liquid chromatography combined with mass spectrometry (HILIC-UPLC-ESI-MS) was employed for accurate glycan mass determination and verification of our original structural assignments. The technology development and advancement in modern MS with high resolution, high selectivity and mass accuracy has enabled it to continue to play a vital role in mapping glycoprotein glycosylation and defining glycan structures (Harvey 2005; North et al. 2009; Zaia 2010). There are several formats for MS that differ in their mode of ionization of analytes. Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS) is rapid and widely used to gain information of molecular mass. Permethylation is the most popular derivatization method, as it enhances ionization efficiency of glycans, stabilizes the labile sialic acid moieties and enables the detection of both neutral and acidic glycans in positive ion mode. MALDI-TOF-MS can be used in conjunction with exoglycosidase sequential digestion for the structural and linkage analysis of glycans (Ressom et al. 2008; Morelle et al. 2009; Gao et al. 2016; Rebello et al. 2020).

We used ESI-MS (Harvey 2000; Balaguer and Neususs 2006; Grunwald-Gruber et al. 2017) to profile the intact glycans without dissociation of fragile acidic groups. This is considerably gentler than the alternative of MALDI-TOF-MS (Banerjee and Mazumdar 2012). Additionally, ESI-MS is directly compatible with online chromatography (Zaia 2009; Zaia 2010; Dreyfuss et al. 2011), improving the ability to detect low abundant glycans or analysis of unseparated mixtures. In our current study, hydrophilic interaction UPLC was coupled to ESI-MS (HILIC-UPLC-ESI-MS) for glycan fragmentation and accurate glycan molecular mass confirmation. HILIC has the advantage that retention times vary predictably according to the glycan composition.

Computational modeling and molecular dynamic (MD) simulation has gradually developed to become a complementary tool to experimental methods. This is because the role of MD simulation is not just limited to the interpretation of experimental results but is also used predictively, leading to the integrated computational and experimental studies on the structure–function relationship of glycans (Duffy and Rudd 2015; Badgett et al. 2018; Woods 2018; Park et al. 2019; Walsh et al. 2019). In addition to the database, we have included a chart that shows the incremental GU values that arise through the addition of specific monosaccharides to a core glycan structure. This will facilitate the prediction of the GU values and identification of unassigned glycan structures based on the glycan core structure and the incremental GU values of specific monosaccharide.

The focus of the current research is to lay a preliminary foundation of an InstantPC glycan GU database, which will benefit from data input from future research. This is because the number of investigated glycoproteins in the current research is limited, and there is considerable value in expanding the potential glycoprotein list and glycan structures. We believe that the InstantPC-labeled glycan GU database in this paper will provide an expandable reference source for the glycans we have analyzed as well as for the addition of new data in the future. This will be of great value in the application of the rapid workflow we have developed for glycan analysis by InstantPC.

Continued support and maintenance of the database may involve four important inputs, including literatures reviews, further experimental determinations, MD simulations and continuous future community input. The database will enhance the InstantPC-based platform for high-throughput and high-sensitivity glycan profiling, with potential use

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in various applications, including (1) biopharmaceutical development; (2) cell recognition, interaction, trafficking and response; (3) glycan-based disease biomarker discovery and (4) glycan-based therapeutic strategies, etc. The constructed InstantPC-labeled glycan GU database is intended to serve as a solid foundation and framework for future applications and advances in the field of glycomics or glycoscience.

In conclusion, our previous study, we presented an innovative streamline 96-well-plate-based high-throughput and high-sensitivity platform for glycan profiling, which holds great potential to be used as sensitive and reliable strategies for high-throughput glycomics analysis. However, unlike the conventional 2-AB based workflow, the limited availability of InstantPC-labeled glycan standards and the lack of an extensive InstantPC glycan GU library limit the value of the new glycan profiling platform. To address this limitation, we have constructed a detailed InstantPC glycans GU database by investigating a variety of glycoproteins from various sources, including human serum, monoclonal antibody (EG2-hFc) produced from CHO cells, lactoferrin from bovine milk, fetuin from fetal bovine serum, transferrin human (partially iron-saturated), ovalbumin from chicken egg white and ribonuclease A type III-A from bovine pancreas. The constructed GU value database for InstantPC-labeled glycans contains the GU values for both high- and ultra-performance H(U)PLC-HILIC-FLD. The reported InstantPC-labeled glycans GU database is robust with good reproducibility and applicability. It is supplemented with a chart of incremental GU values that can be applied to predict unassigned glycan structures. The GU value database for InstantPC-labeled glycans will lay a solid foundation to promote the application of this glycan profiling platform for the benefit of glycan-related biopharmaceutical development and biomedical research.

Materials and methods

Materials and reagents

The AdvanceBio Gly-X *N*-Glycan Prep with InstantPC kit, 96-ct (GX96-IPC), Sialidase S (*Streptococcus pneumoniae*, GK80021), Sialidase A (*Arthrobacter ureafaciens*, GK80040), β (1–3,4)-galactosidase (bovine testis, GKX5013), β (1–4)-galactosidase (*S. pneumoniae*, GKX5014), β-*N*-acetylhexosaminidase (*S. pneumoniae*, GK80050) and α (1-2,3,6)-Mannosidase (jack bean, GKX-5010) were donated by Agilent Technologies (Santa Clara, California, USA). α (1-2,4,6)-fucosidase O (Omnitrophica bacterium, P0749) was purchased from New England Biolabs (Ipswich, Massachusetts, USA). Human serum from human male AB plasma (H4522), IgG from human serum (I4506), lactoferrin from bovine milk (L9507), fetuin from fetal bovine serum (F2379), transferrin human, partially iron saturated (T8158), ovalbumin from chicken egg white (A5503) and ribonuclease A type III-A from bovine (R5125) were purchased from Sigma Aldrich (St. Louis, Missouri, USA). EG2-hFc was produced from CHO cells and was purified by protein A affinity chromatography as described previously (Xie et al. 2021). HPLC-grade acetonitrile was purchased from Sigma (St. Louis, Missouri, USA), and Milli-Q water was used in all preparations. All the common chemicals were purchased from Sigma Aldrich (St. Louis, Missouri, USA).

Preparation of InstantPC-labeled glycans

The preparation of InstantPC-labeled glycans of human serum or other glycoproteins was carried out according to the manufacturer's instruction (Agilent AdvanceBio Gly-X *N*-Glycan Prep with InstantPC kit). Human serum (1 μL) or glycoprotein (40 μg) samples were diluted with a suitable amount of 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid buffer (50 mM, pH 8.0) to make a final volume of 20 μL. Gly-X denaturant (2 μL) was added to the 20 μL of glycoprotein solution, mixed thoroughly and incubated at 90°C for 3 min. After leaving at room temperature for 2 min, 2 μL of *N*-glycanase working solution (*N*-glycanase: Gly-X digestion buffer 1:1 [v/v]) was added, mixed thoroughly and incubated at 50°C for 5 min. InstantPC dye solution was prepared by dissolving one vial of InstantPC dye with 150 μL of the accompanying solvent and mixed well. The InstantPC dye solution (5 μL) was added to the above prepared sample and incubated at 50°C for 1 min. The load/wash solution (150 μL of 2.5% formic acid/97.5% acetonitrile) was added to each sample, and then the entire sample (179 μL) was transferred to each well of the Gly-X Clean-up plate containing 400 μL of the load/wash solution. After passing the solution through the clean-up plate by applying a vacuum, samples were washed with 600 μL of the load/wash solution for three times. InstantPC-labeled glycans were eluted with 100 μL of Gly-X InstantPC eluent (160 mM ammonium formate/10% [v/v] acetonitrile, pH 4.4). The collected InstantPC-labeled glycan solutions were analyzed immediately without further treatment, or were alternatively stored at −20°C for future analysis.

Fractionation of InstantPC-labeled glycans by WAX-HILIC-FLD

For glycoproteins with complex glycan profiles (serum and bovine fetuin), the previously prepared InstantPC-labeled glycans were first fractionated by WAX-HILIC with FLD (WAX-HILIC-FLD) using the 1260 Infinity HPLC system (Agilent Technologies, Santa Clara, California, USA) equipped with GlycanPac™ AXH-1 Column (2.1 mm × 150 mm, 1.9 μm, Cat NO: 082472, Thermo Fisher Scientific, Waltham, Massachusetts, USA) under the control of OpenLAB CDS ChemStation software (Agilent Technologies, Santa Clara, California, USA). The HPLC system consists of a quaternary solvent pump, autosampler and a fluorescence detector. The detector for InstantPC was set with excitation and emission wavelengths at 285 nm and 345 nm, respectively. The above prepared InstantPC-labeled glycan samples were injected at a volume of 5 μL without any further treatment prior to injection. The InstantPC-labeled glycans were fractionated with 100 mM ammonium formate (pH 4.4) as solvent A, H_2O as solvent B and acetonitrile as solvent C. Solvent B (H_2O) was kept constant at 20% throughout the whole fractionation process. After extensive system equilibrium (A:B:C, 10%:20%:70% [v/v/v]), the fractionation was carried out by a linear gradient of 70–60% of solvent C (v/v) in 0–8 min, followed by 60–50% solvent C (v/v) in 8–18 min. Subsequently, a linear gradient of 50–70% solvent C (v/v) in 18–20 min was used and maintained for another 5 min to equilibrate the system for the next fractionation cycle. Samples were maintained at 5°C before injection and the column temperature was set as 30°C. The flow rate was set at 0.3 mL/min throughout the whole fractionation process. The collected InstantPC-labeled

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glycan fractions were dried by speedVac Vacuum Concentrator (Thermo Fisher Scientific, Waltham, Massachusetts, USA) and were stored at −20°C for future analysis. The InstantPC-labeled glycan fractions collected from the equivalent of 40 μg glycoprotein or 1 μL human serum were adequate for the subsequent exoglycosidase sequential digestion.

Exoglycosidase sequential digestion of InstantPC-labeled glycans

Exoglycosidase sequential digestion of the InstantPC-labeled glycans or fractions collected from WAX-HILIC-FLD fractionation was accomplished by the addition of 2.0 μL of α (2–3,6,8,9) sialidase S, sialidase A, β (1–4)-galactosidase, β-*N*-acetyl-hexosaminidase, α (1-2,3,6)-mannosidase and α (1-2,4,6)-fucosidase O to 10 μL of InstantPC-labeled glycans or fractions containing reaction mixture (released from equivalent of 4 μg glycoprotein). For multiple exoglycosidase sequential digestion, 10 × reaction buffer of 0.5 M ammonium acetate (pH 5.5) was utilized. The reactions were incubated at 37°C for 24 h without shaking. The enzymatic digested InstantPC-labeled glycan samples were injected into the chromatographic system at a volume of 1 μL without any prior treatment.

InstantPC-labeled glycan profiling by HILIC-FLD

The profiles of InstantPC-labeled glycans, fractions or exoglycosidase sequential digestion mixtures were determined by HILIC-FLD using three different liquid chromatography systems. The first was an Acquity I class UPLC system equipped with Acquity UPLC Glycan BEH Amide Column (130 Å, 1.7 μm, 2.1 mm × 150 mm, SKU: 186004742) under the control of Empower software (Waters Corporation, Milford, Massachusetts, USA). The second was a 1290 Infinity II Bio UPLC system equipped with AdvanceBio Glycan Map column (300 Å, 1.8 μm, 2.1 mm × 150 mm, P.N.: 859700-913) under the control of OpenLab CDS software (Agilent Technologies, Santa Clara, California, USA). The third was a 1260 Infinity HPLC system equipped with AdvanceBio Glycan Mapping column (120 Å, 2.7 μm, 2.1 mm × 150 mm, Part NO: 683775–913) under the control of ChemStation software (Agilent Technologies, Santa Clara, California, USA). The detector for InstantPC was set with excitation and emission wavelengths at 285 nm and 345 nm, respectively. The InstantPC-labeled glycans, fractions and exoglycosidase sequential digestion samples were injected at a volume of 1 μL without any prior treatment. The InstantPC glycans were separated with 50 mM ammonium formate (pH 4.4) as solvent A and acetonitrile as solvent B. After initial system equilibrium for 1.5 min with 27% of 50 mM ammonium formate (pH 4.4) and 73% acetonitrile (v/v) at a flow rate of 0.5 mL/min, the separation was carried out by a linear gradient of 73–62% of acetonitrile (v/v) at a flow rate of 0.5 mL/min in 40 min, followed by a linear gradient of 62–53% acetonitrile (v/v) at a flow rate of 0.5 mL/min in 12 min. Complete system equilibrium under 27% of 50 mM ammonium formate (pH 4.4) and 73% acetonitrile (v/v) at a flow rate of 0.5 mL/min for another 15 min is necessary to ensure good chromatographic reproducibility. Samples were maintained at 5°C before injection, and the separation column temperature was 60°C. The above chromatographic conditions were utilized for three LC systems and columns and all the investigated glycoproteins without further optimization. All three systems were routinely calibrated using Agilent AdvanceBio InstantPC Maltodextrin ladder (GKPC-503). The correlation between GU value and chromatographic retention time T (min) was fitted to a fifth order polynomial function to obtain the standard curve.

Confirmation of InstantPC-labeled glycans mass by HILIC-UPLC-ESI-MS

The accurate molecular mass of InstantPC-labeled glycans were verified by HILIC-UPLC-ESI-MS using Q Exactive Plus (Thermo Fisher Scientific, Waltham, Massachusetts, USA) equipped with Acquity UPLC Glycan BEH Amide Column (130 Å, 1.7 μm, 1.0 mm × 150 mm, SKU: 186004850, Waters Corporation, Milford, Massachusetts, USA). Samples were maintained at 5°C before injection, and the separation temperature was 60°C. The InstantPC-labeled glycans were injected at a volume of 10 μL. The UPLC detector was set with excitation and emission wavelengths at 285 nm and 345 nm for InstantPC, respectively. The glycans were separated with 50 mM ammonium formate (pH 4.4) as solvent A and acetonitrile as solvent B. After initial system equilibrium for 1.5 min with 27% of 50 mM ammonium formate (pH 4.4) and 73% acetonitrile (v/v) at a flow rate of 0.15 mL/min, the separation method was carried out by a linear gradient of 73–62% acetonitrile (v/v) at a flow rate of 0.15 mL/min in 40 min, followed by a linear gradient of 62–53% acetonitrile (v/v) at a flow rate of 0.15 mL/min in 12 min. The system was equilibrated between samples by 73% acetonitrile (v/v) at a flow rate of 0.15 mL/min for 15 min to ensure reproducibility. The running condition for MS was positive mode, spray voltage of 3.40 kV, capillary temperature of 320°C, aux gas heater temperature of 300°C, sheath and sweep gas flow rate of 30 L/h and 10 L/h, respectively, scan range of 450–2500 m/z and resolution of 70,000.

Data processing and analysis

The chromatographic glycan peaks from the HILIC-FLD analysis were processed with the built-in software of either Empower or OpenLab CDS ChemStation using an automated method with a traditional integration. Individual glycan peaks were analyzed on the basis of the correlation between measured retention times and GU values generated from the fifth order polynomial standard calibration curve against the AdvanceBio InstantPC Maltodextrin ladder under identical conditions. The HILIC-UPLC-ESI-MS spectra were processed and visualized by Xcalibur software (Thermo Fisher Scientific, Waltham, Massachusetts, USA). The glycan structures and fragments were assigned with the aid of GlycoWorkbench software (Ceroni et al. 2008).

Construction of InstantPC-labeled glycans GU database

The InstantPC-labeled glycans from various glycoproteins (human serum, ovalbumin from chicken egg white, lactoferrin from bovine milk, fetuin from fetal bovine serum, transferrin human partially iron, ribonuclease A type III-A from bovine pancreas and monoclonal antibody of EG2-hFc produced from CHO cells) were profiled by HILIC-FLD. WAX-HILIC-FLD fractionation was employed for glycoproteins with complex glycan profiles. The glycan structural assignments were verified by further exoglycosidase sequential digestion. The accurate molecular mass for each glycans was confirmed

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by HILIC-UPLC-ESI-MS with the glycan molecular weight addition of InstantPC at 261.1477. The average GU value for all measurements of a given InstantPC-labeled glycan from different sources was used in the database. The InstantPC-labeled glycans from polyclonal antibody of human IgG was profiled subsequently for the verification of the constructed InstantPC-labeled glycans GU database.

Supplementary data

Supplementary data are available at *Glycobiology* online.

Authors' contributions

YX: conceptualization, methodology, investigation, writing, review and editing. MB: validation, writing, review and editing.

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Conflict of interest statement

YX was supported as a postdoctoral researcher in NIBRT by Agilent Technologies.

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